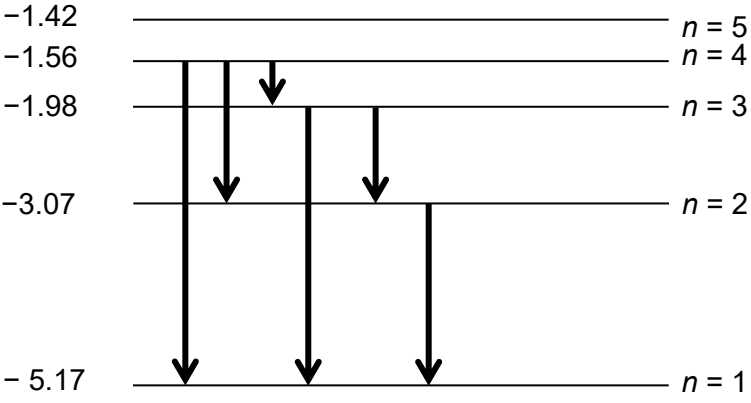
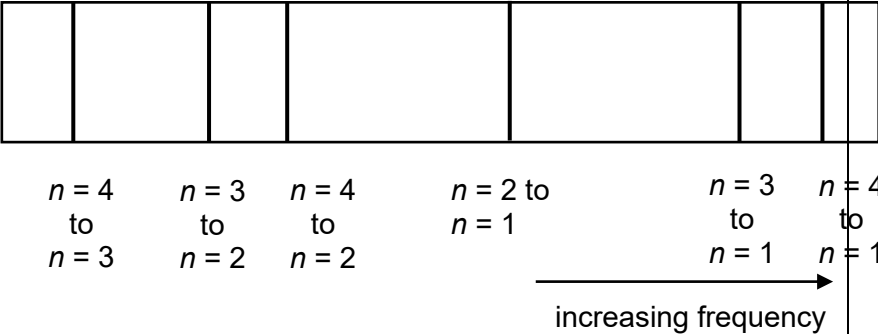


No.	Solution	Remarks
6(a)(i)	If all the electron's energy is absorbed, the electron could reach $-5.17 + 3.7 = -1.47$ eV level. Since there are no such energy level, the electron can only excite to -1.56 eV or $n = 4$ level. energy / eV  Diagram showing energy levels (eV) and transitions: $n = 5$: -1.42 eV $n = 4$: -1.56 eV $n = 3$: -1.98 eV $n = 2$: -3.07 eV $n = 1$: -5.17 eV Transitions shown: $n = 5 \rightarrow n = 1$ $n = 4 \rightarrow n = 1$ $n = 3 \rightarrow n = 1$ $n = 3 \rightarrow n = 2$ $n = 4 \rightarrow n = 2$ $n = 5 \rightarrow n = 2$	[1] for transitions to $n = 1$ [1] for transitions to $n = 2$ and $n = 3$
6(a)(ii)	 Diagram showing spectral lines and transitions: $n = 4$ to $n = 3$ $n = 3$ to $n = 2$ $n = 4$ to $n = 2$ $n = 2$ to $n = 1$ $n = 3$ to $n = 1$ $n = 4$ to $n = 1$ increasing frequency [1] for 3 spectral lines to the left and 2 spectral lines to the right [1] for correct spacing for the extreme 2 spectral lines on the right	
6(a)(iii)	If the electron absorbed the incoming photon, there is <u>no energy levels for it to excite to</u>. As such, there will be <u>no emission lines observed</u>.	[1]
6(b)(i)	The X-ray tube is evacuated so that the electrons emitted from the filament can move towards the metal target without colliding with air molecules and getting scattered.	[1]
6(b)(ii)	A photon is a quantum of electromagnetic radiation having a discrete amount of energy $E = hf$ where h is the Planck constant and f the frequency of radiation.	[1]
6(b)(iii)	From Fig. 6.3, $\lambda_{\min} = 0.40 \times 10^{-11}$ m ,	

No.	Solution	Remarks
	$E = \frac{hc}{\lambda_{\min}}$ $= \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{(0.40 \times 10^{-11})}$ $= 4.97 \times 10^{-14} \text{ J}$	[1] correct substitution [1] correct answer
6(b)(iv)	$qV = 4.97 \times 10^{-14}$ $(1.60 \times 10^{-19})V = 4.97 \times 10^{-14}$ $V = 310.6$ $\approx 310 \text{ kV}$	[1] substitution
7(a)(i)	The engine takes in air at a rate of 1200 kg s^{-1}	